

Digital strategies for the social sciences – 9

Clustering

Introduction

About the final project:

- Don't forget to ask a research question (a 'research' Q)
- Don't forget to i) explain the method and ii) justify its use

Deadline: June 10th, 23h59.

Office hours with Mael

Clustering

‘Grouping individuals together based on similar properties’

Clustering

‘Grouping individuals together based on similar properties’

An essential skill (both qualitative and quantitative)

- Preparation of the data for other methods
- Vast data sets, need to uncover patterns
- Clustering can be used with other methods (create variables)

A non-trivial problem

Clustering

‘Grouping individuals together based on similar properties’

An essential skill

A non-trivial problem

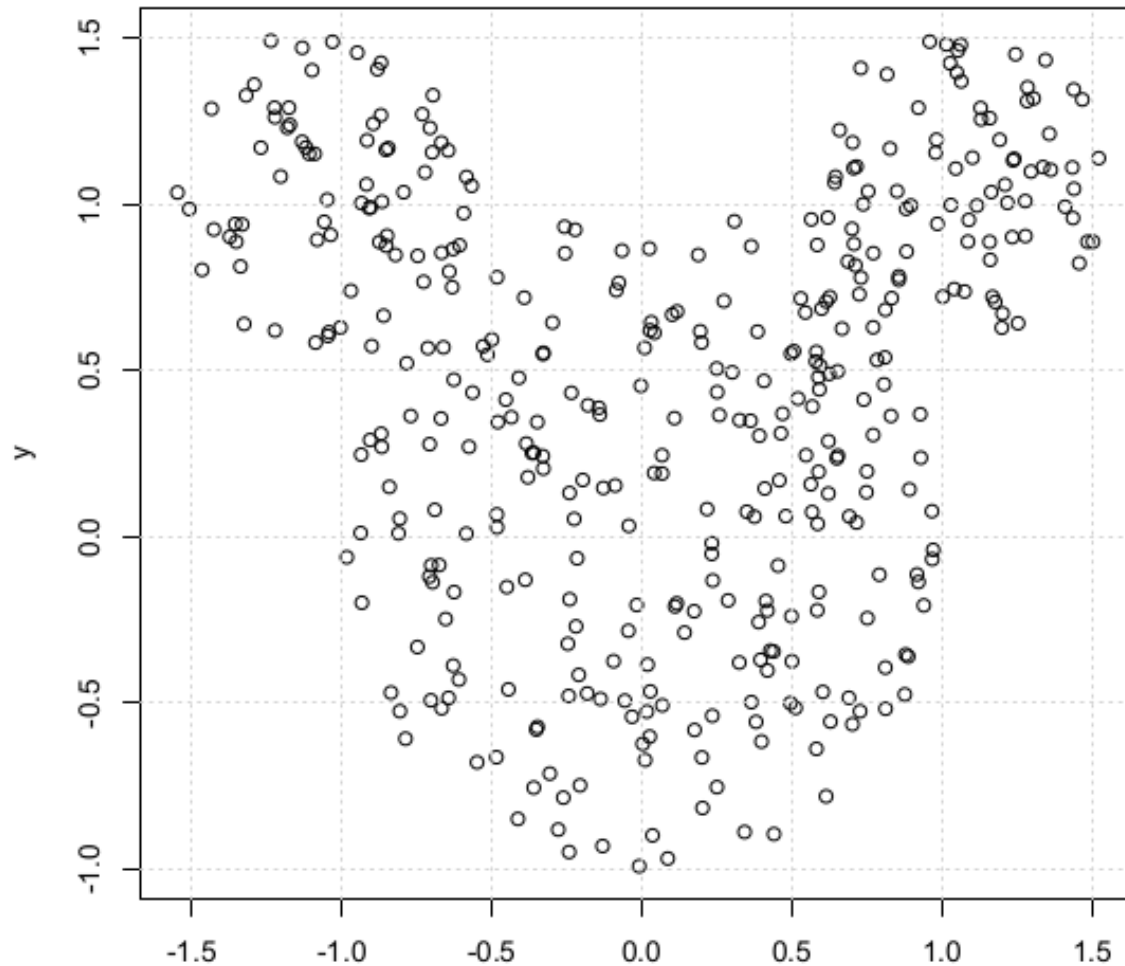
Sociologically

How to group people ?

Clustering

'Grouping individuals together based on similar properties'

How would you cluster me?



Courtesy of J. Boelaert



Clustering

‘Grouping individuals together based on similar properties’

An essential skill

A non-trivial problem

- Sociologically
 - Mathematically
- A potentially NP-hard problem

Clustering

‘Grouping individuals together based on similar properties’

An essential skill

A non-trivial problem

Many options, many choices

- No dominant method
- A central approach, but somewhat bad press
‘The Afghanistan of Statistics’

Clustering

‘Grouping individuals together based on similar properties’

Outline

1. Data preparation
2. Clustering algorithms
3. Choose a clustering algorithm
4. Interpret and Evaluate your clustering

Clustering

Data Preparation

Scaling [aka standardization]

- Scaling is an important procedure, esp. if your variables have different units, or very different values.
- Not scaling will give pride of place to some variables, conceal the effects of others. Unless you have a very good reason not to scale, please do.

$$x' = \frac{x - \bar{x}}{\sigma}$$

Clustering

Clustering algorithms

2 mains families

Partitioning algorithms

Hierarchical Clustering Algorithm

Clustering

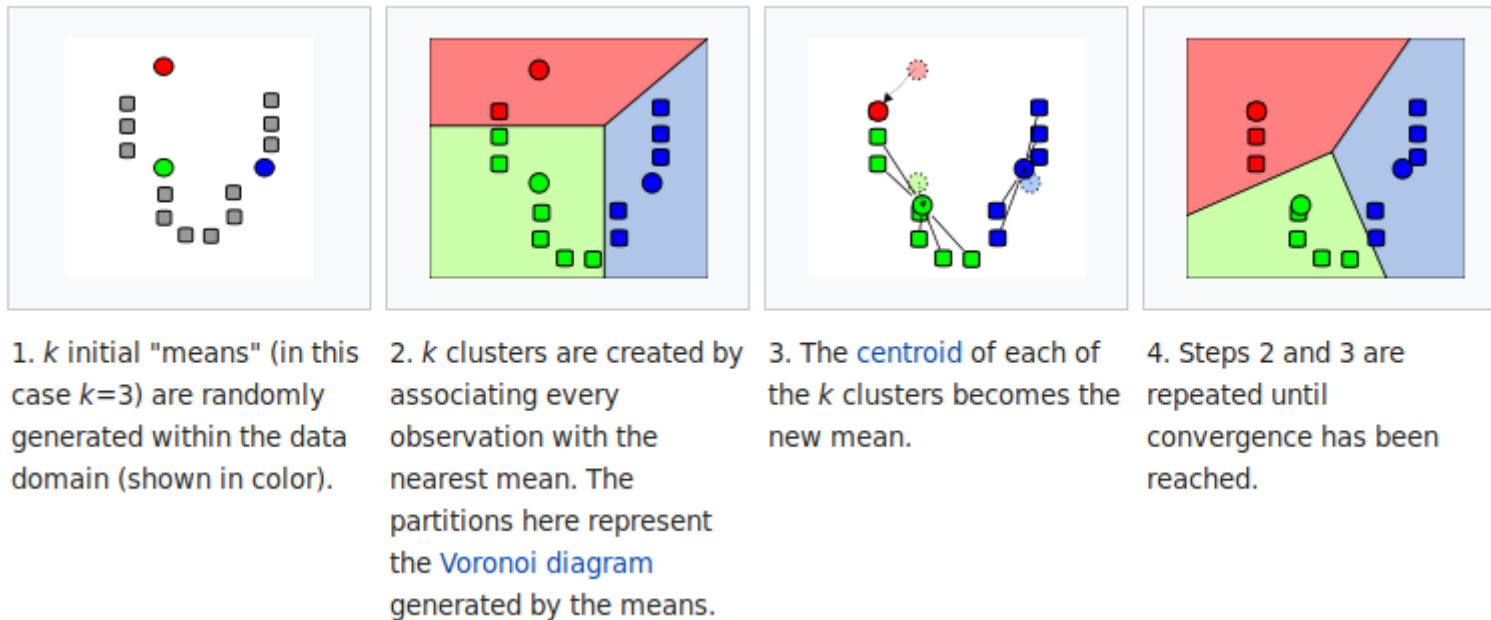
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Partitioning algorithms

- Adjust the partitioning so as to maximize an objective function ("density of the cluster")
- Several tools invented to partition according to a **pre-determined** number of clusters.
- Classic ones: Kmeans, PAM (CLARA)

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Partitioning algorithms

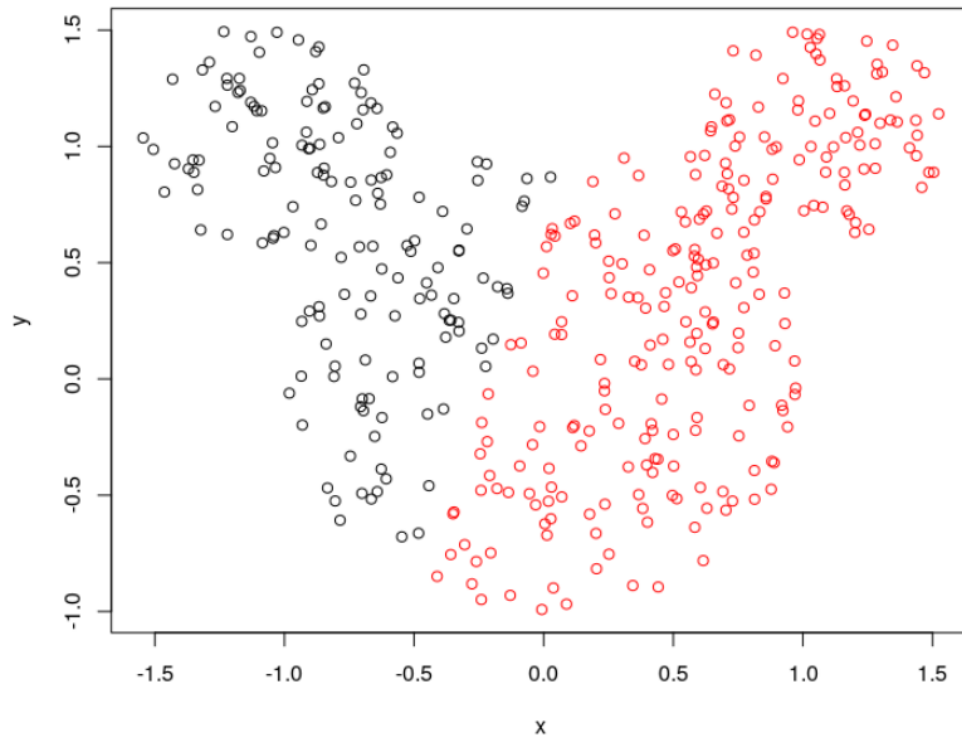
Kmeans intuition

Minimizing the total intra-cluster variation.

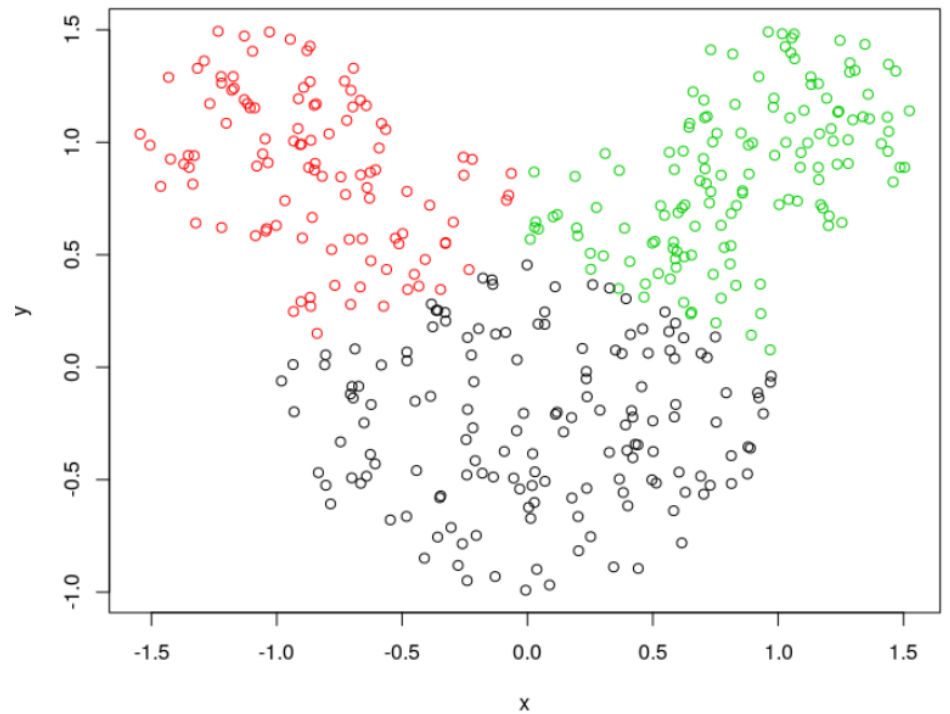
- 1- Choose the k number of clusters you want → k points
- 2- Place the points in the graph
- 3- Assign to each point the closest observations
- 4- Measure the intra-cluster variation
- 5- Move the points to the centroid of the space, go back to 3
- 6- Stop when you have reached the lowest level

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Kmeans-2



Kmeans-3



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Partitioning algorithms: Kmeans

Kmeans is smart, but has issues

- Assumes **knowledge about the # of clusters**
- Sensitive to outliers
- Sensitive to initial positioning of the cluster centers

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Partitioning algorithms: PAM

PAM (also called K-medoids) is derived from Kmeans

- Medoid: central point of the cluster
- Use a similar routine than kmeans, except the strategy to calculate is a bit different (called “swaping”).
- Still, needs to define the number of k before

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Clustering algorithms

2 mains families

Hierarchical Clustering Algorithm

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Hierarchical Clustering Algorithm

- Old approach, that can be greedy (resources)
- Alternative method that does not require to decide ex ante on the number of clusters
- 2 main approaches: divisive, and agglomerative

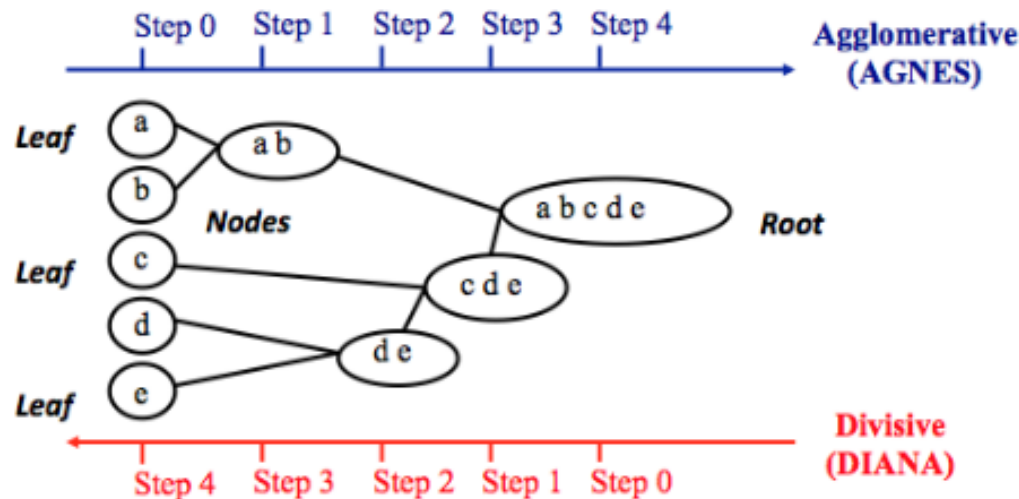
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Hierarchical Clustering Algorithm

- Alternative method that does not require to decide ex ante on the number of clusters
- 2 main approaches: agglomerative (leaves → tree), and divisive (tree → leaves)



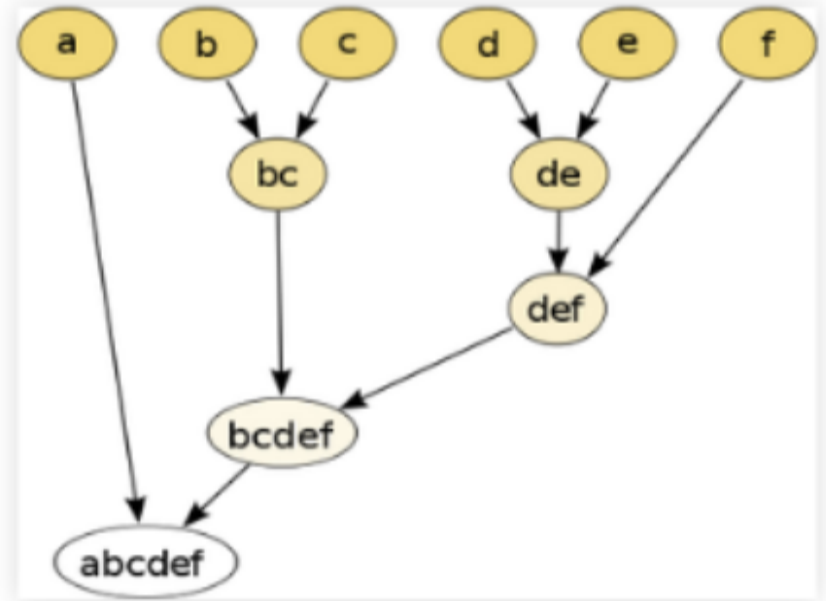
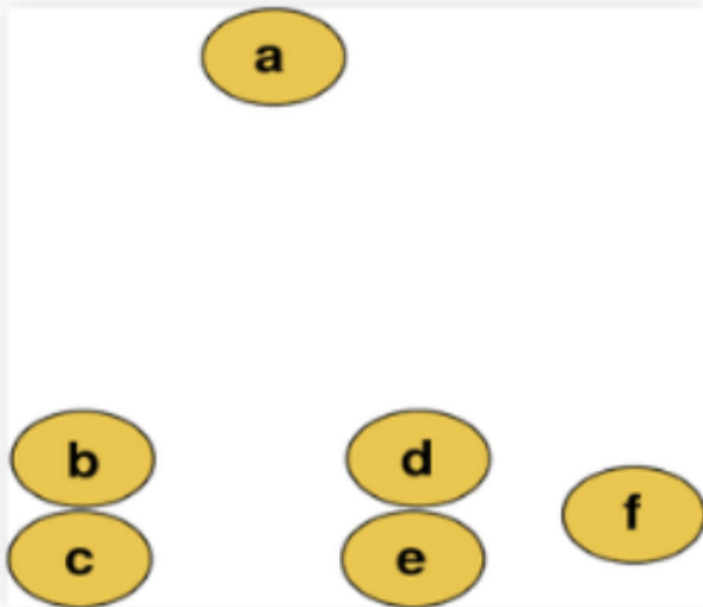
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- Alternative method that does not require to decide ex ante on the number of clusters
- Ex. Agglomerative



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Steps

- Prepare the data (scale?)
- Calculate a distance matrix
- Choose a linkage function
- Determine where to cut, based on the dendrogram

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Important moment 1: Distance matrix

- Euclidean (most often the case, unless good reason)
- Manhattan
- Correlation-based

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Important moment 2: Linkage strategy

- Single: Catches outliers
- Ward: Classic, close to PAM
- Complete: Compact spheres

Name	Criterion	Properties
Ward	Intra-class variance	Compact clusters, spheres
Single linkage	Nearest neighbors	Outlier detection
Complete linkage	Furthest away neighbors	Close to Ward. Good is neatly separated neighbors

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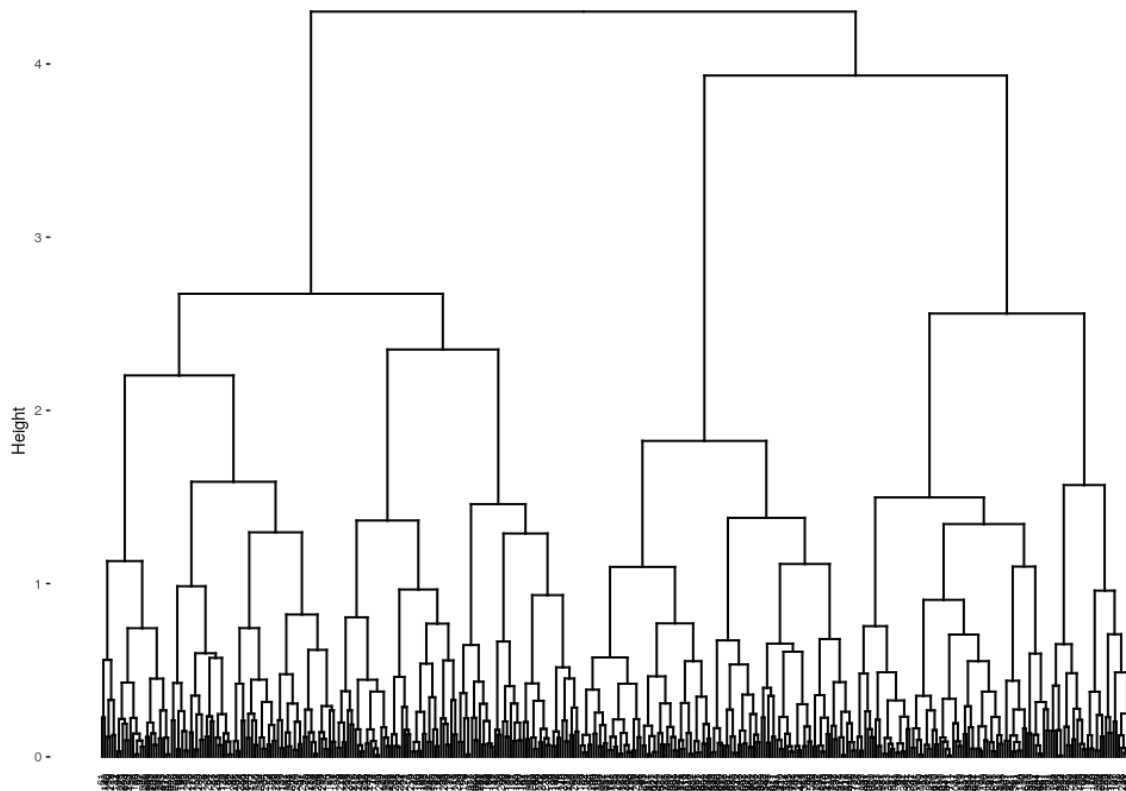
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Important moment 3: Dengrogram

Cluster Dendrogram



Where to cut ?

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Interpreting & Evaluating

Given the number of choices you had to make, it is important to test your clusters. Here again, many options are on the table.

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Interpreting & Evaluating

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Statistics: Silhouettes

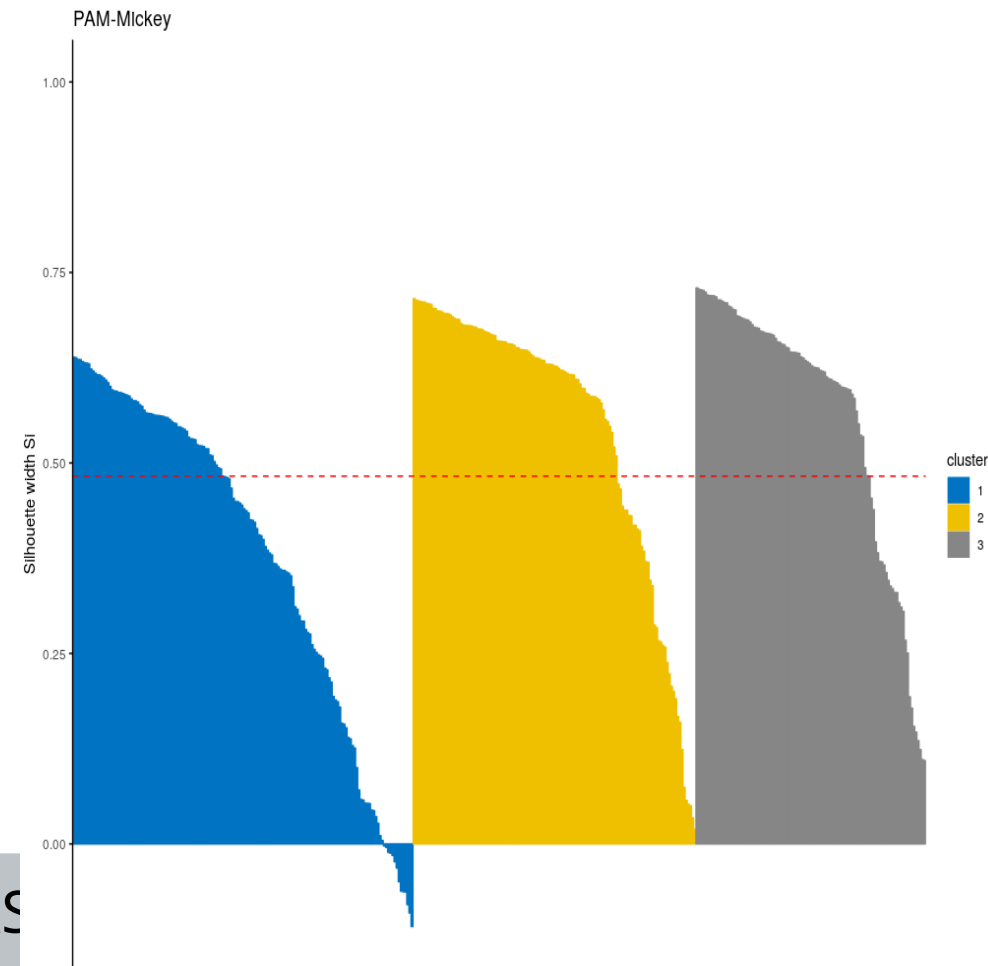
How well is a given individual clusterized ?

Distance between the point and the clusters.

Close to 1 : well-allocated

Close to 0 : bordering two clusters

Negative: ill-clustered



Clustering

Interpreting & Evaluating

Given the number of choices you had to make, it is important to test your clusters. Here again, many options are on the table.

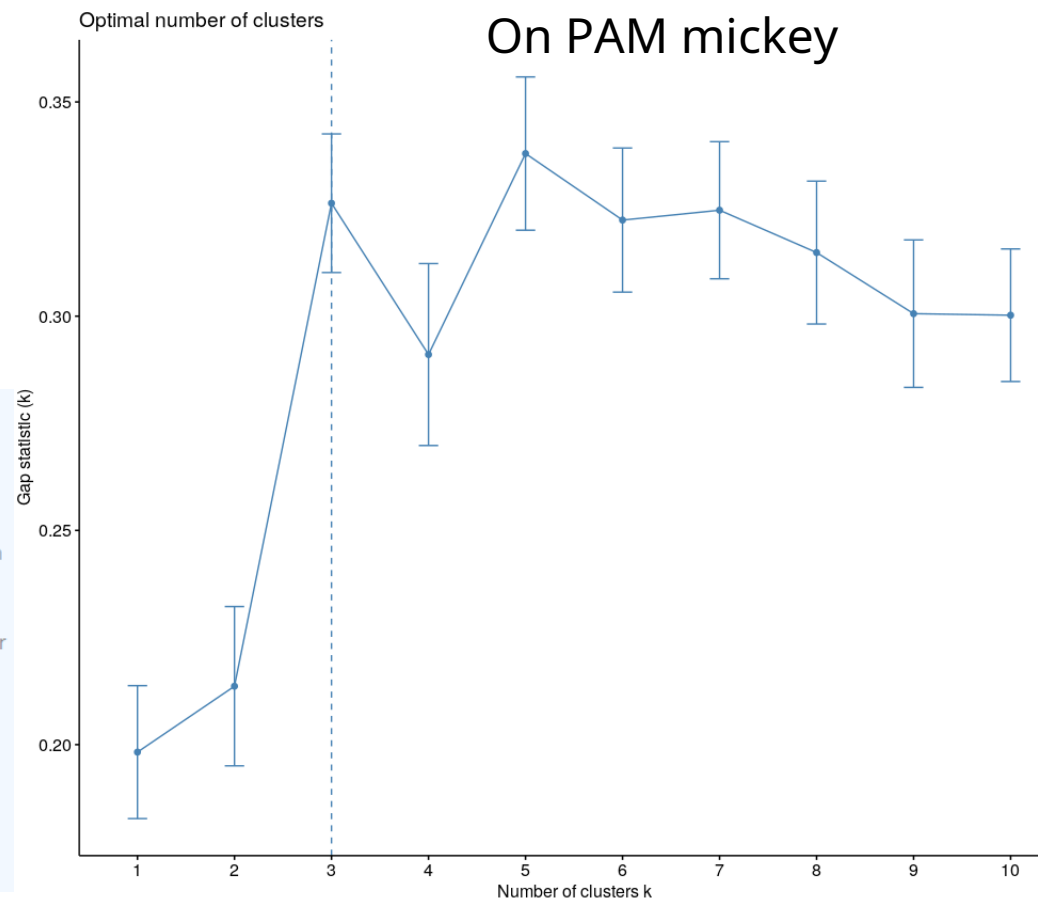
Statistics: Gap

Compares the intra-cluster variance to that of the same amount of clusters, but on randomly distributed data.

(Tibshirani, 2001)

=> Highest value before a gap (drop)

1. Cluster the observed data, varying the number of clusters from $k = 1, \dots, k_{max}$, and compute the corresponding total within intra-cluster variation W_k .
2. Generate B reference data sets with a random uniform distribution. Cluster each of these reference data sets with varying number of clusters $k = 1, \dots, k_{max}$, and compute the corresponding total within intra-cluster variation W_{kb} .
3. Compute the estimated gap statistic as the deviation of the observed W_k value from its expected value W_{kb} under the null hypothesis: $Gap(k) = \frac{1}{B} \sum_{b=1}^B \log(W_{kb}^*) - \log(W_k)$. Compute also the standard deviation of the statistics.
4. Choose the number of clusters as the smallest value of k such that the gap statistic is within one standard deviation of the gap at $k+1$: $Gap(k) \geq Gap(k+1) - s_{k+1}$.



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Interpreting & Evaluating

Given the number of choices you had to make, it is important to test your clusters. Here again, many options are on the table.

Interpreting

Do your cluster make sense ? All of them ?

Look for patterns in the aggregated data

```
> aggregate(iris[,1:4], by=list(pam.iris$clustering), mean)
  Group.1 Sepal.Length Sepal.Width Petal.Length Petal.Width
1      1      5.006000      3.428000      1.462000      0.246000
2      2      6.811111      3.082222      5.553333      1.997778
3      3      5.812727      2.700000      4.376364      1.412727
> aggregate(iris[,1:4], by=list(iris$Species), mean)
  Group.1 Species Sepal.Length Sepal.Width Petal.Length Petal.Width
1      1  setosa      5.006      3.428      1.462      0.246
2      2 versicolor  5.936      2.770      4.260      1.326
3      3  virginica  6.588      2.974      5.552      2.026
```

In this case on iris, for which we know the species

Clustering

Interpreting & Evaluating

Keep in mind: no statistics will tell you the truth with non supervised methods.

They are just indicators to help you decide, to see if you are plausibly right.

Non supervised method are meant to be interpreted, this is what matters.